

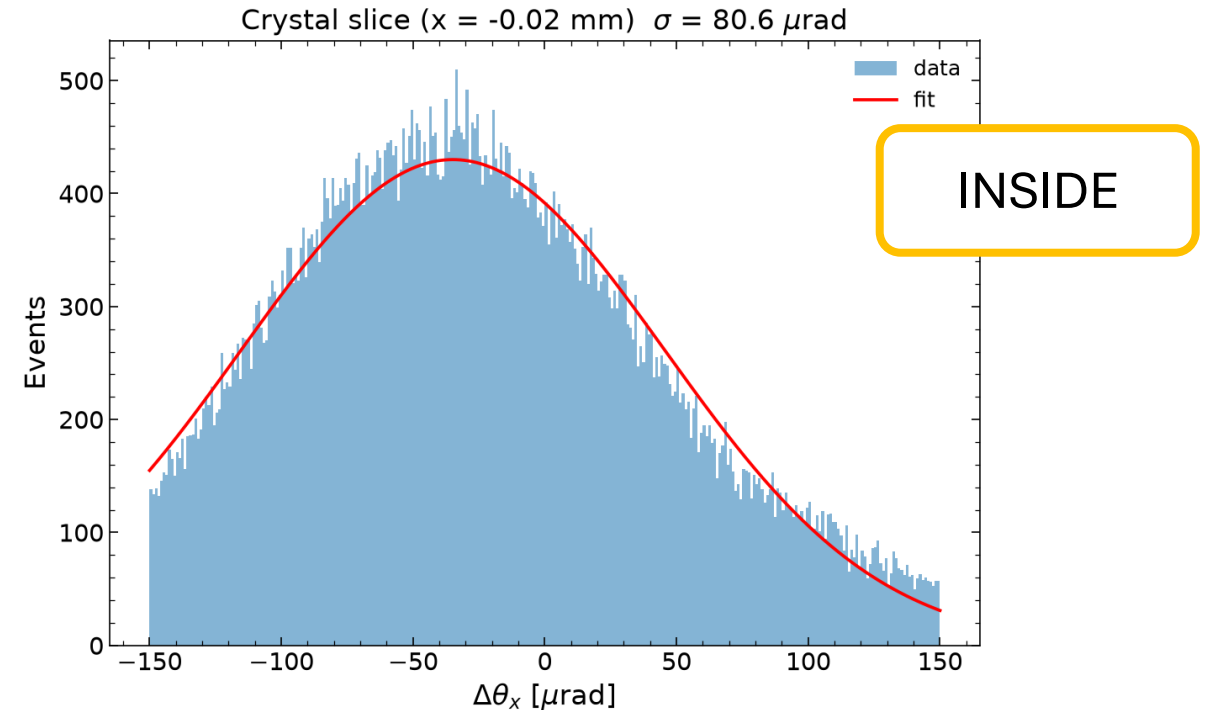
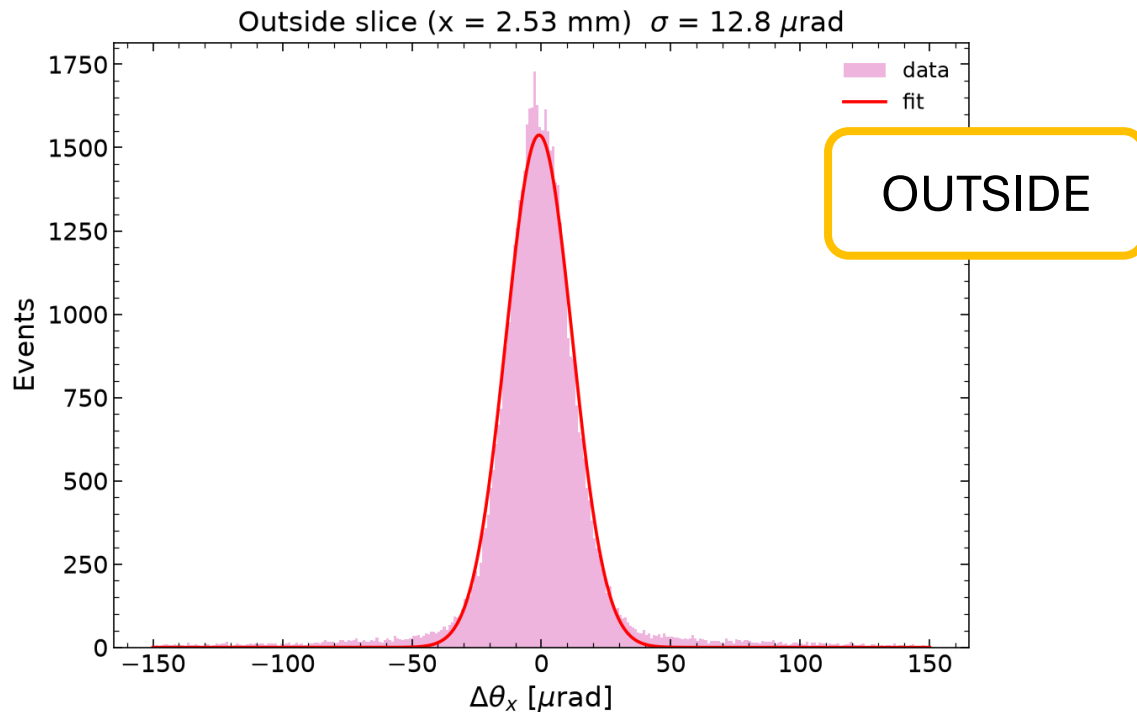
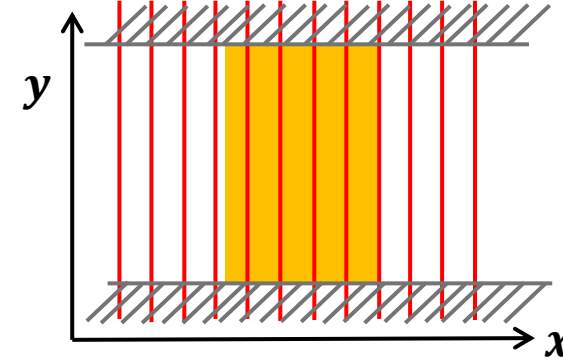
SEPTEMBER 2026 UPDATE
ANNA SOFIA MORO

WHAT'S NEW?

1. Multiple Coulomb Scattering
 - Analysis made for both x and y
 - Clamp material evaluation
2. Analysis of the beam distribution of real data
 - Check the distributions
 - Check the correlation/covariance
3. Simulation
 - Reproduce the real data beam features
 - Channeling efficiency analysis
4. Thesis
 - Possible order of topics in the table of contents

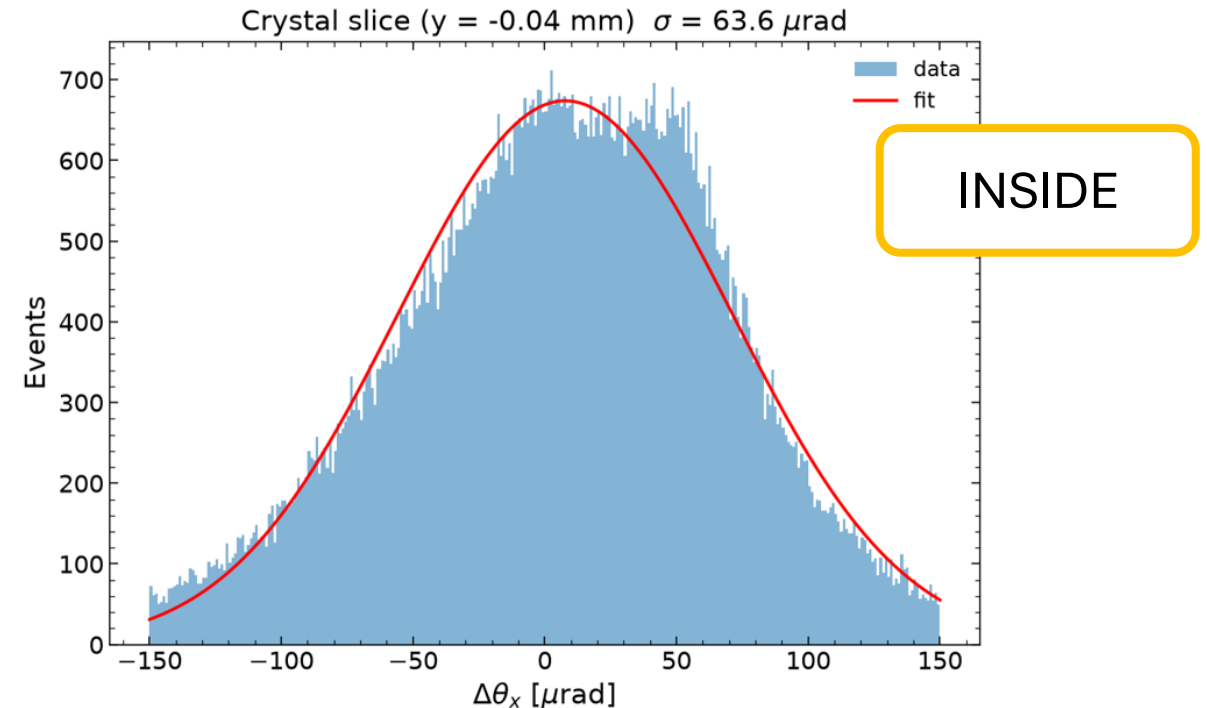
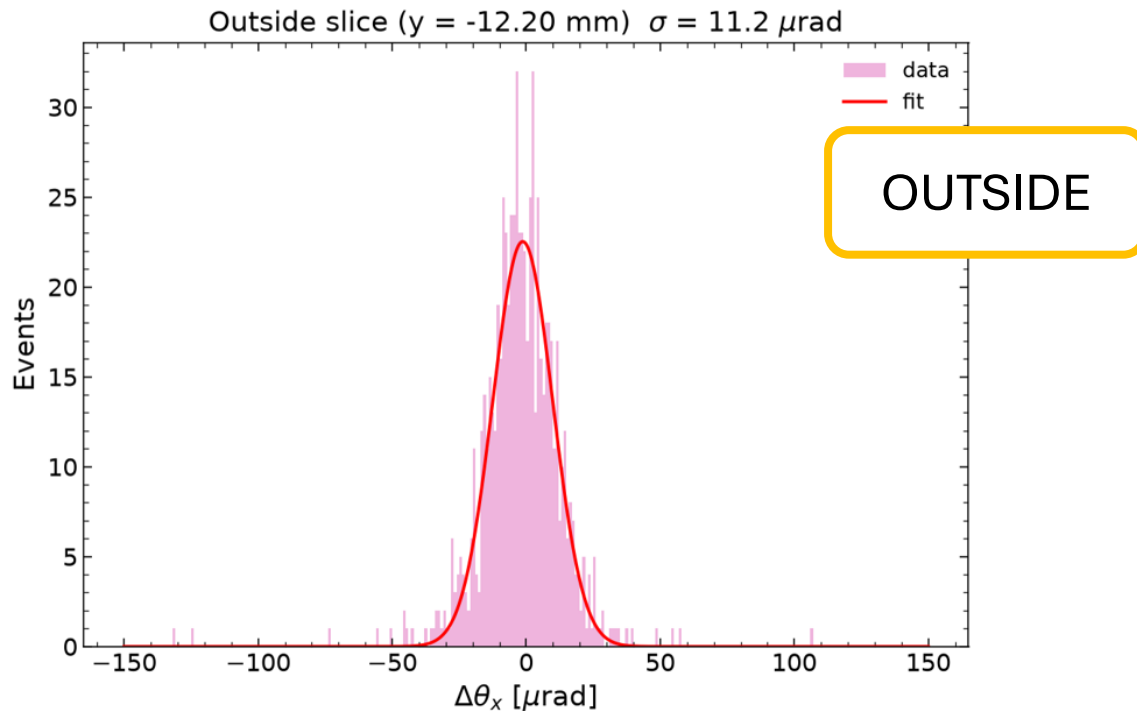
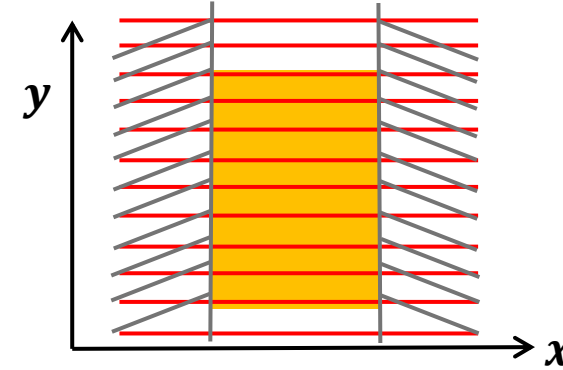
FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

- Divided the crystal into slices along x or y
- Width of the angular distribution σ for each slice



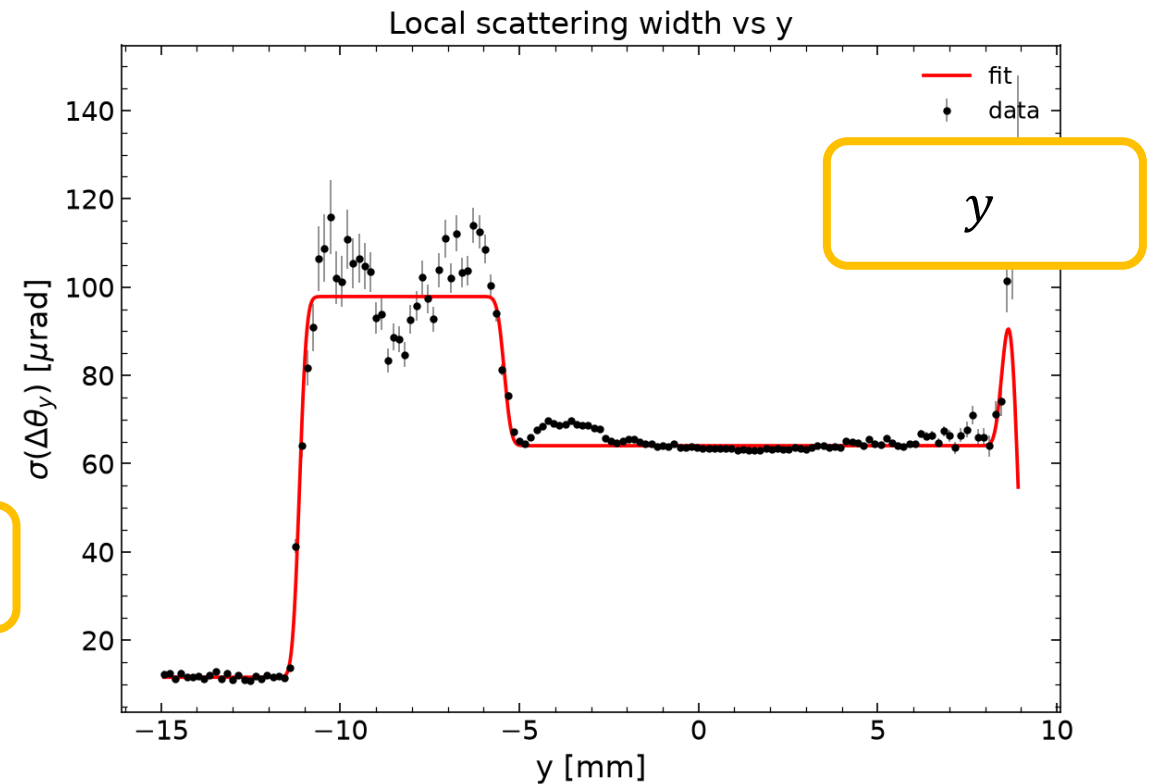
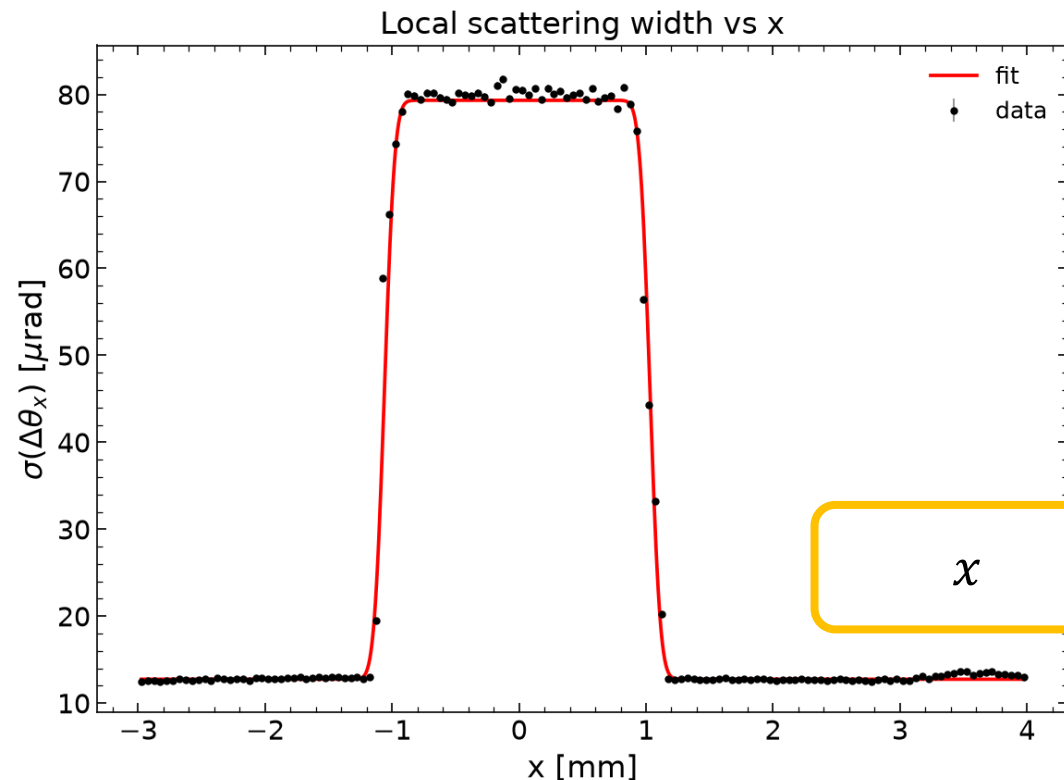
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FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

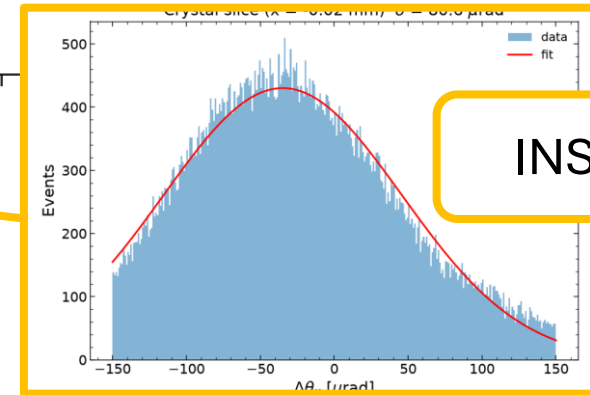
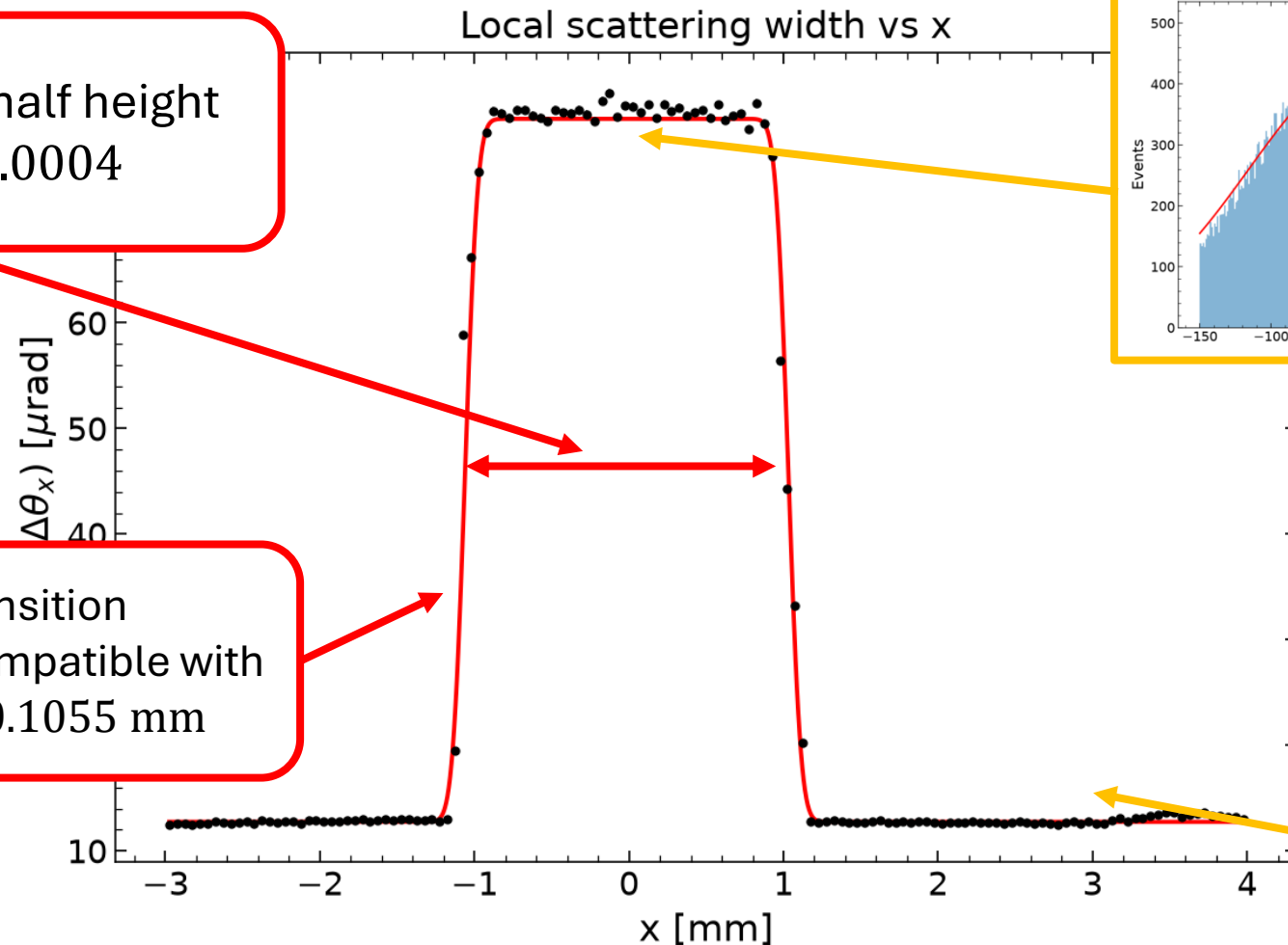
- The extracted gaussian widths σ are plotted against the impact position
- The transition between the outside and inside regions is fitted using a Error function (Erf)



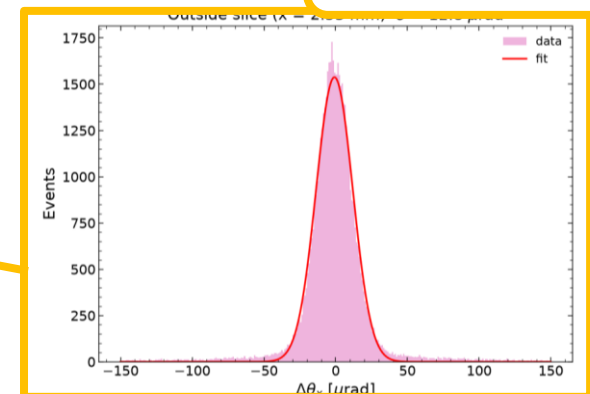
FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

The window @ half height
 2.0911 ± 0.0004

The width transition
 $0.0901(3)$ mm is compatible with
position error ~ 0.1055 mm



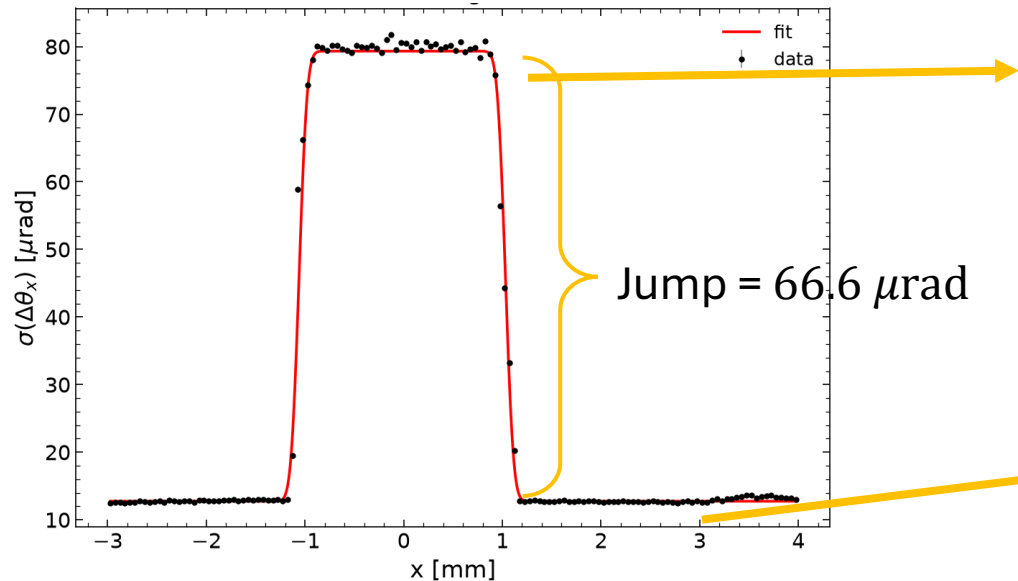
INSIDE



OUTSIDE

FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

$$\langle \theta_{MCS} \rangle = \frac{13.6 \text{ MeV}}{\beta c p} z \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln \left(\frac{x z^2}{X_0 \beta^2} \right) \right] = 66.5 \mu\text{rad}$$



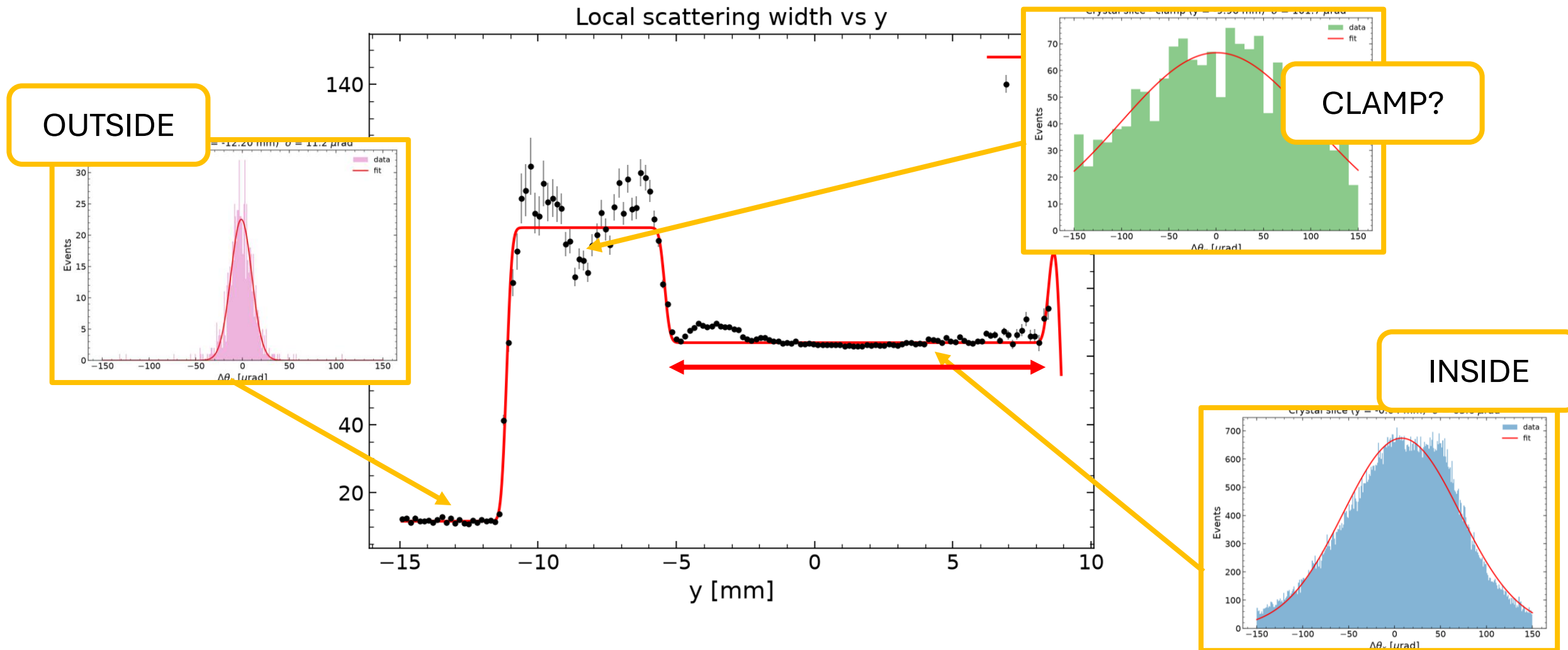
$$\sigma_{meas} = \sqrt{\sigma_{mcs}^2 + \sigma_{track}^2} \sim 79.5 \mu\text{rad}$$

$$\rightarrow \sigma_{mcs} = 78.30(6) \mu\text{rad} \neq \sigma_{mcs}^{th}$$

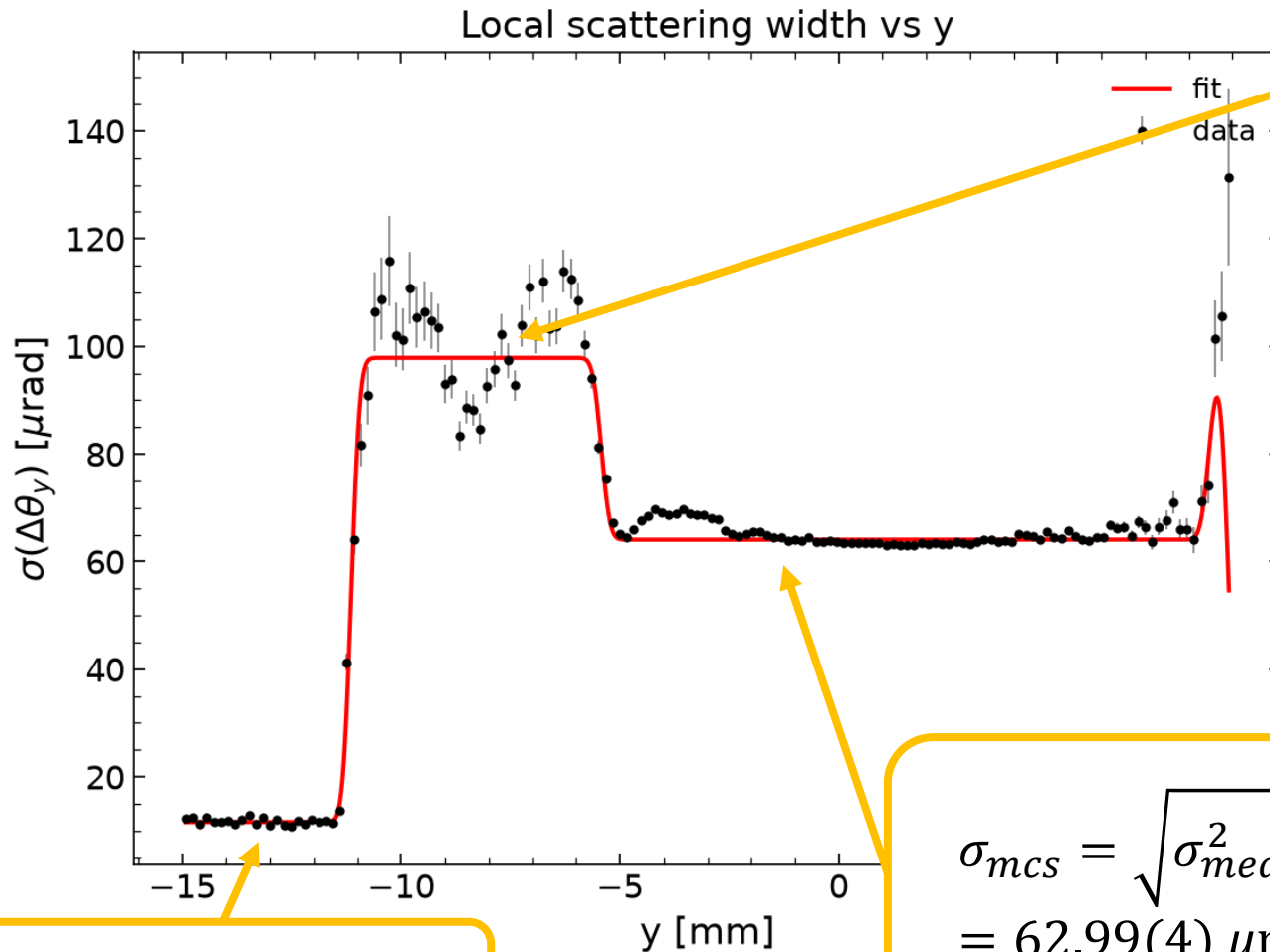
not zero, but $\sim 12.75(1) \mu\text{rad}$
which corresponds to σ_{track}

$$\sigma_{track} = \sqrt{\sigma_{track,in}^2 + \sigma_{track,out}^2} = 12.6 \mu\text{rad}$$

FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING



FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING



CLAMP: which material/metal?
on average $\sim 99.2 \pm 0.3 \mu\text{rad}$

unknown: x, X_0

Between different metallic materials (stainless steel, aluminium, titanium, copper, brass):

→ The only similar is titanium

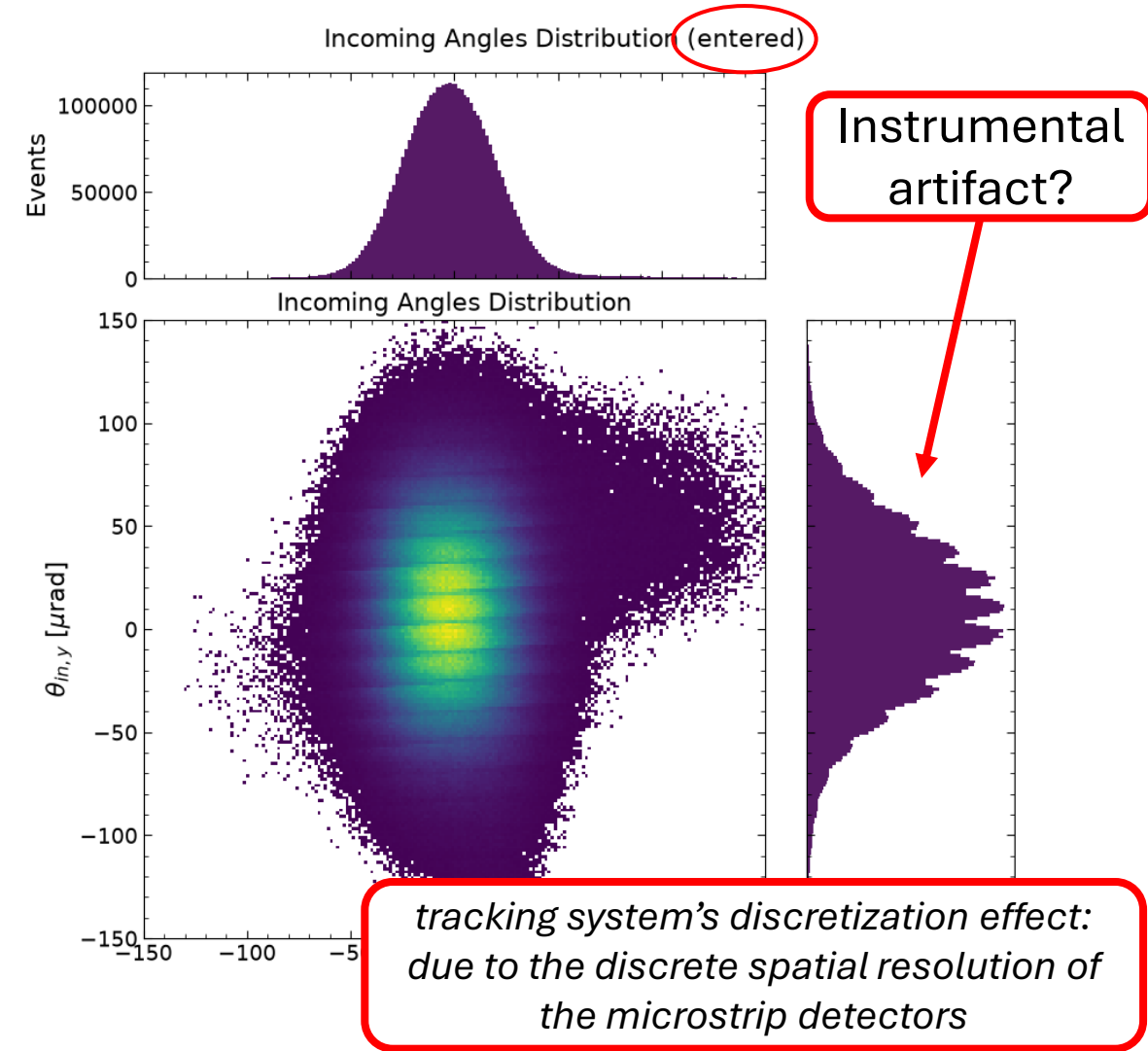
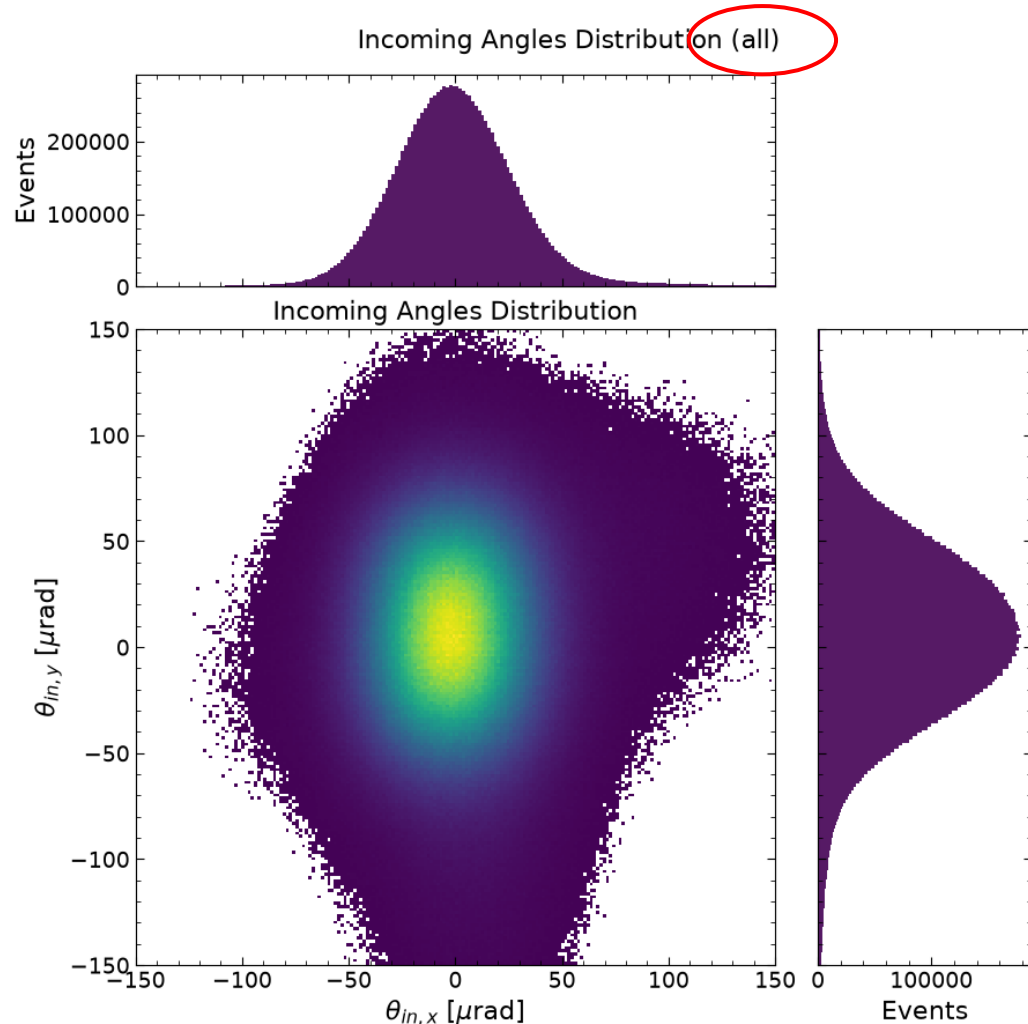
$\theta \sim 110 \mu\text{rad}$

(stainless steel too much scattering, aluminium not enough scattering)

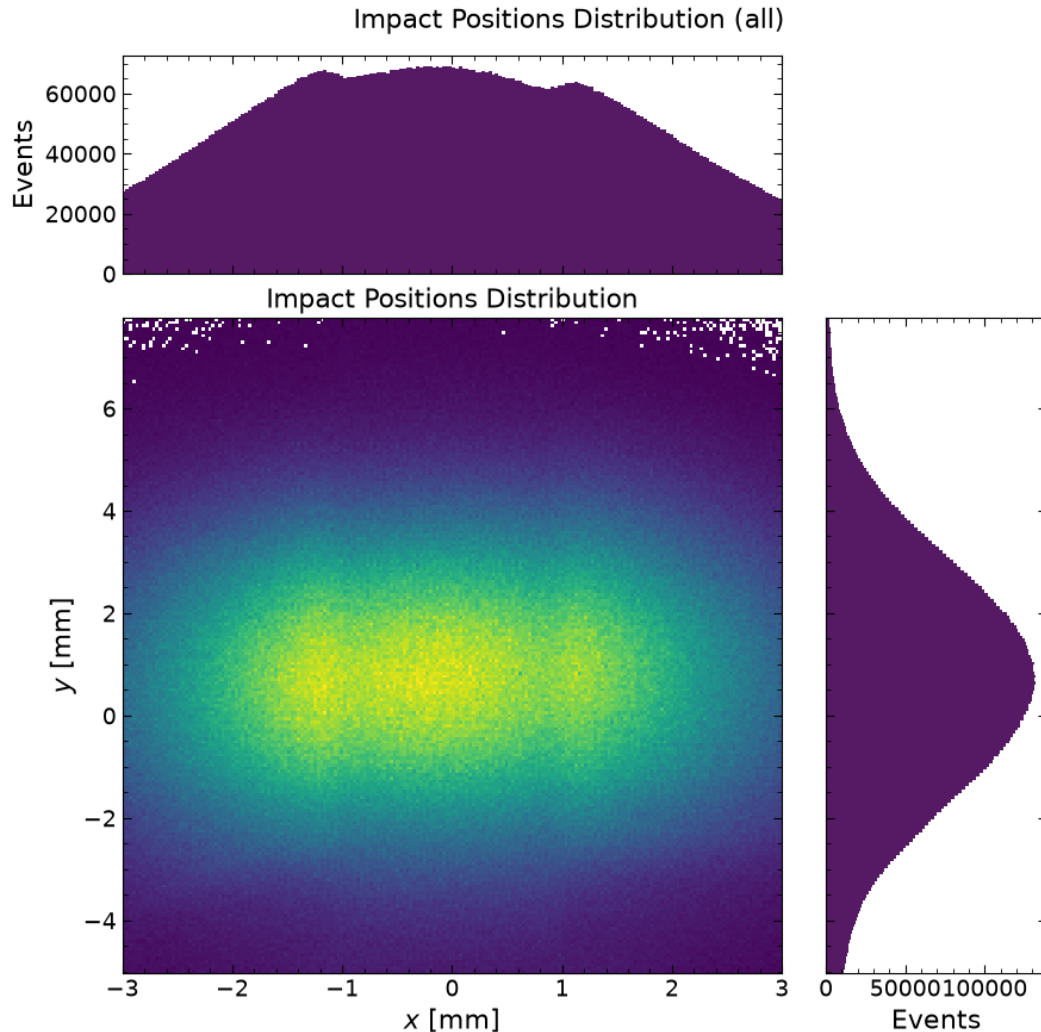
$$\sigma_{bsl} = 11.6(1) \mu\text{rad}$$

$$\sigma_{mcs} = \sqrt{\sigma_{meas}^2 - \sigma_{bsl}^2}$$
$$= 62.99(4) \mu\text{rad}$$

ANALYSIS OF THE BEAM DISTRIBUTION (REAL DATA)



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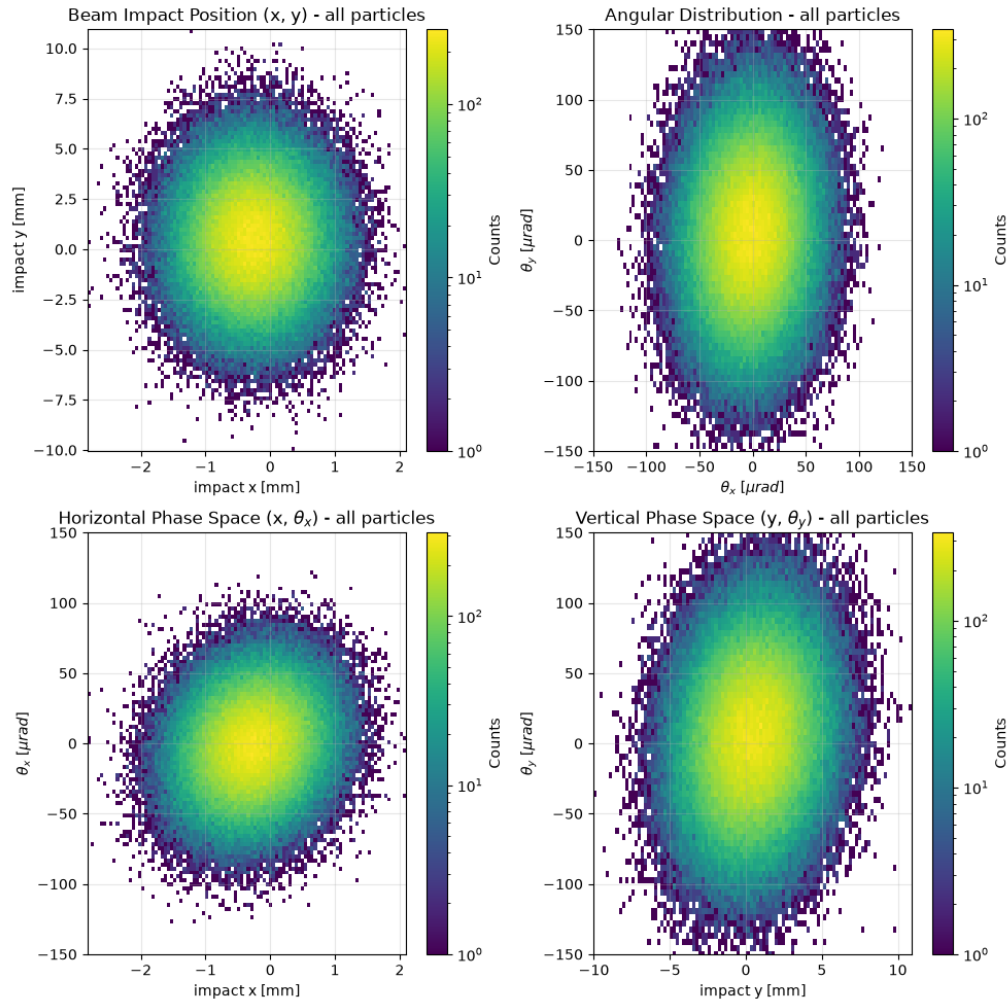
✓ Gaussian fits on the distributions of x, y, θ_x, θ_y

✓ Check the correlation:

Not negligible for $x - \theta_x$, $y - \theta_y$:

	x	y	θ_x	θ_y
x	1	0	0.2	0
y	0	1	0	0.2
θ_x	0.2	0	1	0
θ_y	0	0.2	0	1

SIMULATING THE SHORT CRYSTAL



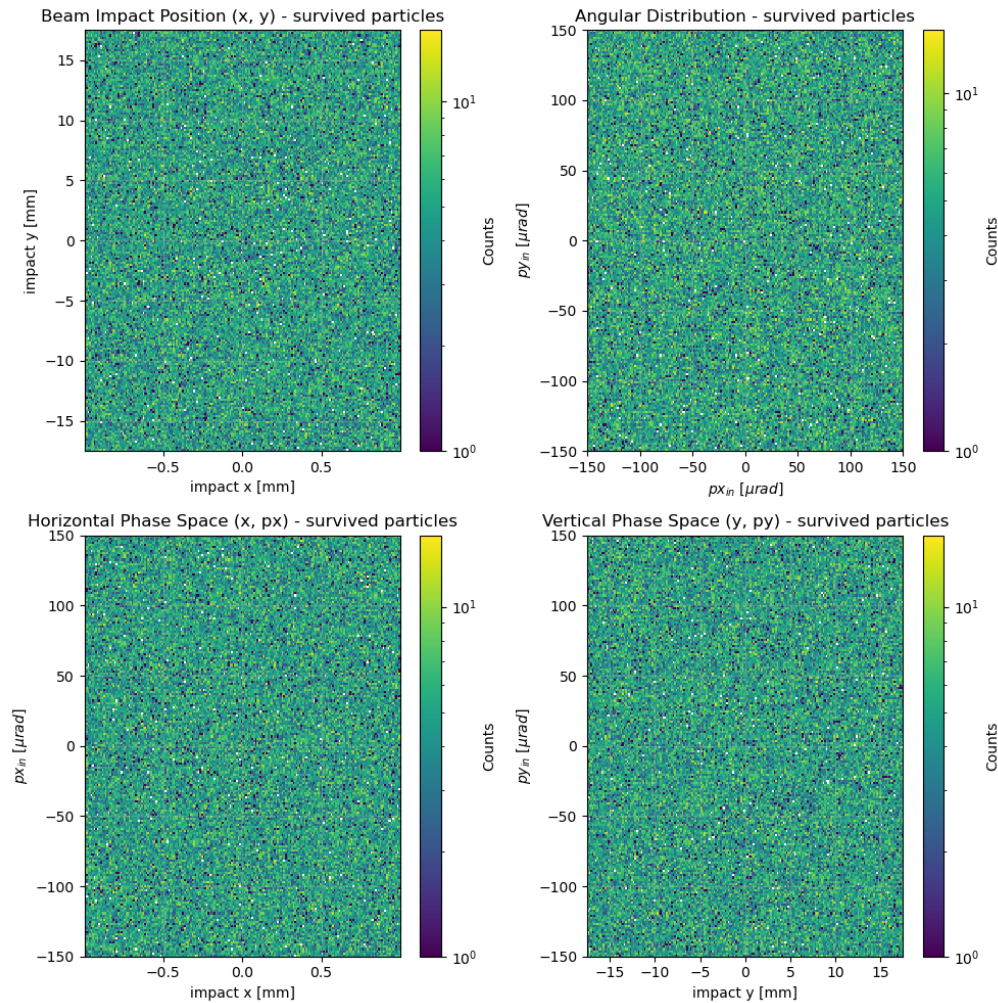
Fitting the 1d marginal distributions and then sampling them independently would not be correct if variables are uncorrelated.

Solution:

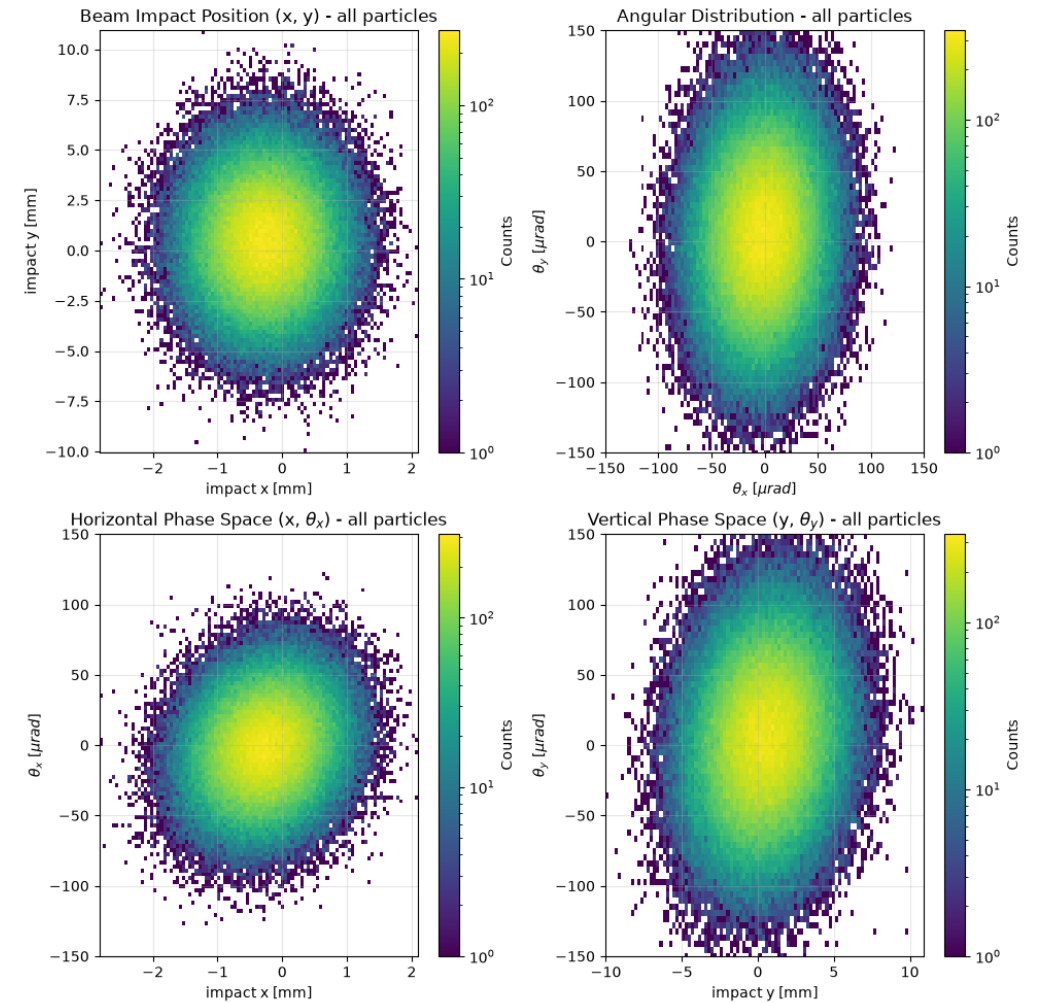
Using a multivariate random normal distribution (thanks to the covariance matrix reproduce the correlation)

SIMULATING THE SHORT CRYSTAL

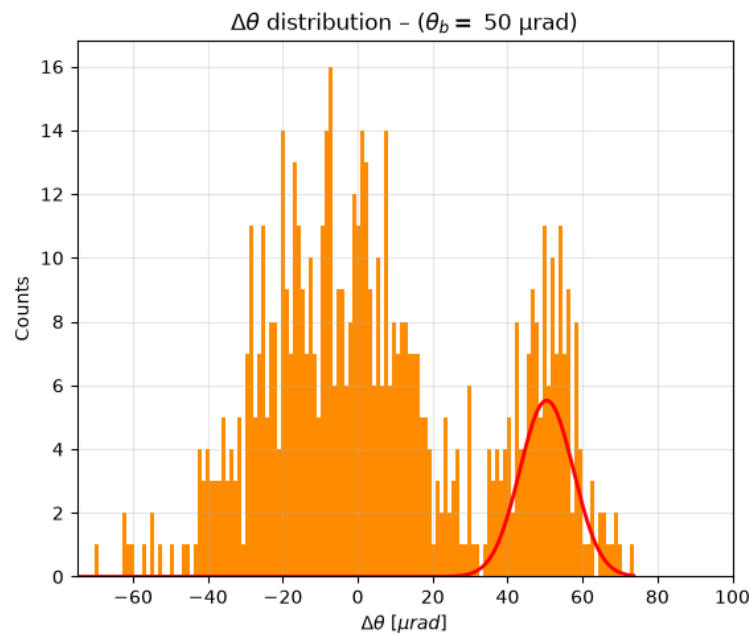
Before: normal distribution



Now: normal distribution



SIMULATING THE SHORT CRYSTAL



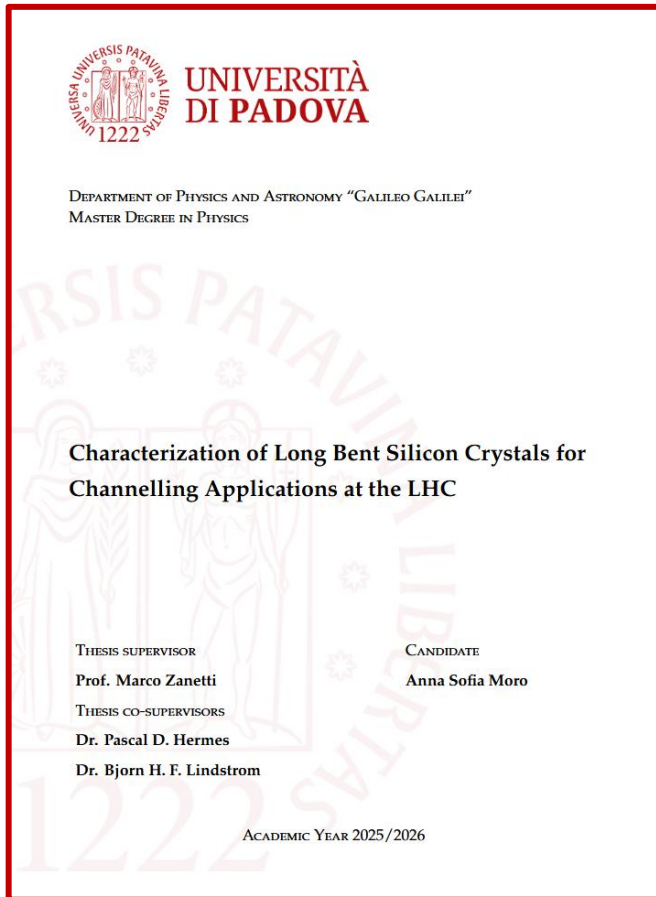
Out of 200k particles only ~ 650 are surviving

But with $\sigma_x = 0.568$ mm and the crystal with 1 mm width, the fraction that should hit the crystal is 92%.

Possible problems?

- Analysing the impact position that the lost particles would have had, they would have both been outside or inside the crystal (they are getting lost also if they would have travel through air)
- Aperture values, energy cut, ...

THESIS: PART I



- Physics background
 - Dipole moment
 - Λ_C^+
- Crystal Channeling
 - In straight/bent crystals
 - Other processes (dechannelling, volume capture, ...)
- Experimental principle: measuring spin precession through bent crystals
- TWOCRIST experiment
- H8 testing